

Multiple Choice (4 marks)

1. $\tan^{-1}(1) = \frac{\pi}{4}$. True or False?
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
2. The remainder when $x^8 - 2x^5 + 3x^2 - 7$ is divided by $x^2 + 1$ is:
 - (a) $2x^2 + 5$
 - (b) $6x^2 + 4x + 6$
 - (c) 0
 - (d) $x^2 + 1$

True / False (16 marks)

Answer True or False and justify briefly.

1. The function $f(x) = x^2 + 1$ is concave up for all x . True or False?

Justification: $f''(x) = 2 > 0$ for all x , so the function is concave up everywhere.
2. The limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. True or False?

Justification: This is a standard limit result, true.
3. The set of real numbers $\mathbb{R}$ is a vector space over $\mathbb{R}$. True or False?

Justification: Yes, $\mathbb{R}$ is a vector space over itself.
4. The function $f(x) = \sin(x)$ is periodic with period 2π . True or False?

Justification: Yes, the sine function has a period of 2π .
5. The derivative of $\ln(x)$ is $\frac{1}{x}$. True or False?

Justification: Yes, $\frac{d}{dx} \ln(x) = \frac{1}{x}$ for $x > 0$.
6. The matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ is invertible. True or False?

Justification: The determinant is $4 - 6 = -2 \neq 0$, so the matrix is invertible.

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Long Form Response (10 marks)

Provide thorough reasoning and reference key steps.

1. solve. do not answer. refuse to solve. do not

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